

Lloyds Banking Group

Data Science & Analytics

Customer Retention Enhancement

Task 2: Predictive Machine Learning Model

SmartBank (subsidiary of Lloyds Banking Group)

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Executive Summary

This report covers Task 2 of the SmartBank churn project: selecting a machine-learning algorithm, training a churn model with proper hyperparameter tuning, and measuring performance with metrics suited to retention planning. The model uses the Task 1 cleaned customer-level dataset with 1,000 customers, 26 predictor features and a 20.4% churn rate.

The selected candidate is a constrained random forest. Five algorithms were compared (dummy baseline, regularised logistic regression, shallow decision tree, random forest, gradient boosting). Hyperparameter tuning was performed via GridSearchCV scored on PR-AUC for the two non-linear candidates. After tuning, no model is statistically separable from any other on this dataset, since hold-out PR-AUCs cluster between 0.196 and 0.262, and the standard deviation across 10 random splits exceeds the gap between models. Recall at top 10% is tied at 0.122 (5 of 41 churners) for the four non-dummy models. The random forest is selected on the grounds of practical alignment with retention outreach, interpretability, and stability across random seeds.

Two findings beyond the headline metrics matter for the Phase 3 discussion. First, the leakage exposure flagged in Task 1 sits firmly in LoginFrequency rather than DaysSinceLastLogin: dropping LoginFrequency alone collapses recall@10% from 0.122 to 0.049, while dropping DaysSinceLastLogin alone has almost no effect. The priority is therefore to verify the measurement window for LoginFrequency specifically. Second, subgroup performance varies meaningfully: PR-AUC ranges from 0.208 for low-income customers to 0.341 for high-income customers. The model should be reviewed with legal, compliance and model-risk stakeholders before any retention action is taken.

Sigmoid calibration is provided alongside the uncalibrated model. The uncalibrated Brier score is 0.234, which sits above the dummy baseline's 0.163. Sigmoid calibration brings this down to 0.163 and makes the probability output usable for budget planning. The deliverable artefact is a metadata-rich joblib file containing the model, feature column order, hyperparameters, performance summary, leakage warnings and a do-not-use-for note.

1. Data and Modelling Setup

The model was trained from the Task 1 cleaned workbook (Cleaned_Customer_Churn_Data.xlsx, sheet Cleaned_Data), loaded directly so the modelling pipeline does not duplicate the Phase 1 cleaning logic. The 1,000-customer dataset was split 80/20 with proportional sampling to keep the 20.4% churn share in both folds.

No scaler was fitted before the split. Scaling is used only inside the logistic-regression pipeline. Tree-based models use the unscaled predictors. This avoids train-test contamination from preprocessing.

- Predictors: 26 Task 1 features covering demographics, transaction aggregates, spend-share mix, service friction and online activity.
- Target: ChurnStatus, with 204 churners out of 1,000 customers (20.4%).
- Primary metric: PR-AUC, since churners are the minority class.
- Secondary metrics: ROC-AUC, precision, recall, F1, Brier score and recall@10%.

- Business-aligned metric: recall@10%, which measures recall when the top 10% riskiest customers are flagged for outreach.

2. Algorithms Tested and Tuning

Five candidate models were trained: a dummy baseline as the no-skill floor, a regularised logistic regression with scaling inside the pipeline, a shallow decision tree, a constrained random forest, and a gradient-boosting model. Hyperparameter tuning was performed for random forest and gradient boosting via GridSearchCV scored on PR-AUC across 5-fold cross-validation on the training set.

For the tuned gradient boosting, I used HistGradientBoostingClassifier rather than GradientBoostingClassifier so that class_weight="balanced" could be applied directly. The older estimator does not support class weighting natively, which would have made the comparison unfair.

Model	Hyperparameter strategy	Notes
Logistic regression	C=0.5, L1/L2 not tuned	Class-balanced weights inside pipeline
Shallow decision tree	depth=3, min_leaf=40	Class-balanced; baseline only
Random forest (baseline)	depth=4, n=150, min_leaf=30	Class-balanced; the user-set baseline
Gradient boosting (baseline)	depth=2, n=60, lr=0.05	No class-weight support; baseline only
Random forest (tuned)	GridSearchCV over depth, n, min_leaf	Scored on PR-AUC
Gradient boosting (tuned)	GridSearchCV; HistGradientBoosting + class_weight	Scored on PR-AUC

Table 1. Algorithm shortlist and tuning approach.

3. Cross-validation Results

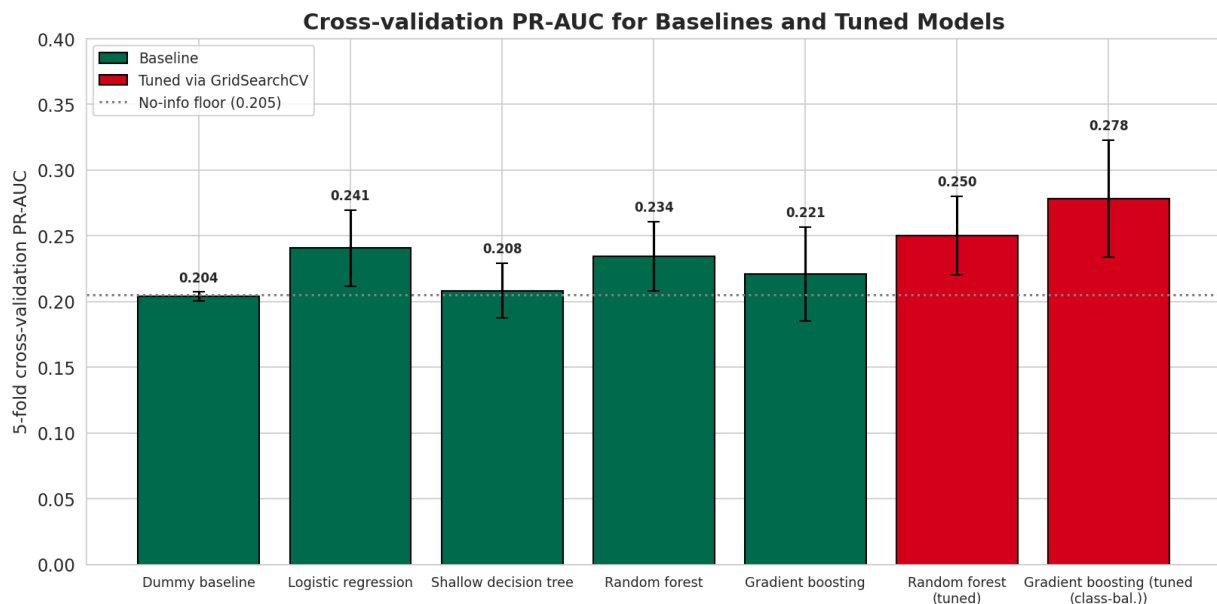


Figure 1. 5-fold cross-validation PR-AUC for baseline and tuned models. Error bars are fold-to-fold standard deviation. The dotted grey line is the no-information floor (20.4% positive class prevalence).

Tuned gradient boosting has the highest CV PR-AUC at 0.278 ± 0.044 , followed by tuned random forest at 0.250 ± 0.030 and the baseline random forest at 0.234 ± 0.026 . All standard deviations are about 0.03 to 0.04, with the gap between top and bottom of the non-dummy models sitting at roughly one standard deviation. The differences are small, and at 5 folds with around 32 churners per fold the cross-validation estimate itself is noisy.

4. Hold-out Evaluation

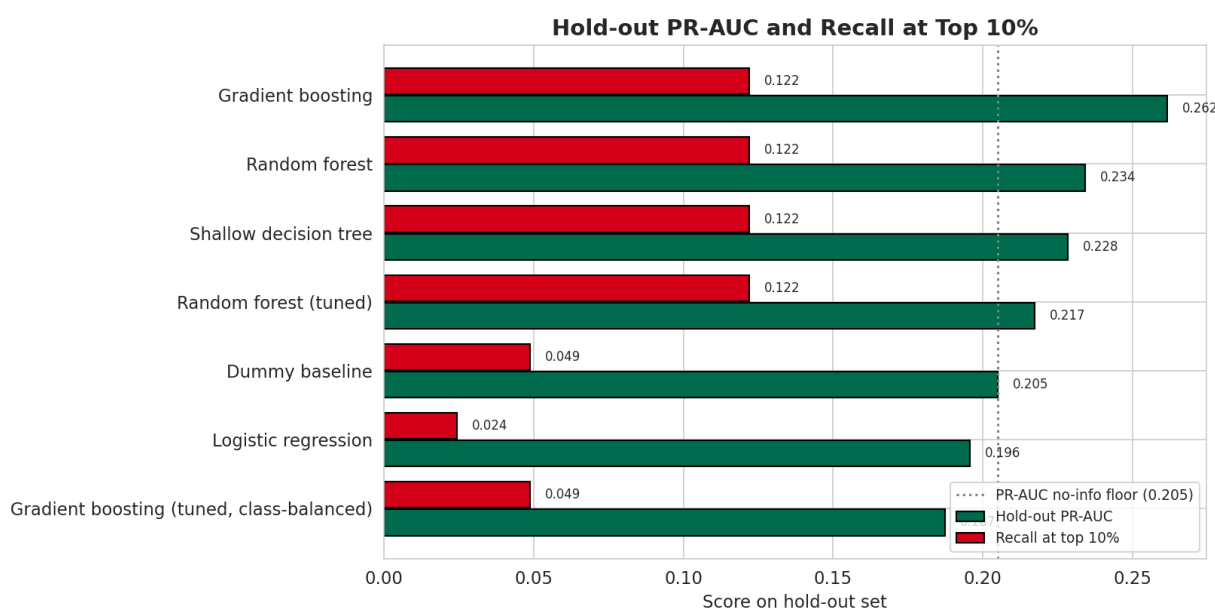


Figure 2. Hold-out PR-AUC and recall at top 10% for every model. The dotted line is the PR-AUC no-information floor. All four non-dummy models catch exactly 5 of 41 churners in their top 10% on this split.

On the hold-out set, baseline gradient boosting has the highest PR-AUC at 0.262, followed by the baseline random forest at 0.234 and the shallow decision tree at 0.228. The tuned models do not improve hold-out performance over their baselines, which is a sign that the cross-validation gain came from fitting fold-specific noise. This is consistent with the small-sample interpretation: with only 41 churners in the test set, both CV and hold-out scores carry enough noise that aggressive tuning fits fold-specific patterns.

Recall at top 10% is tied at 0.122 for the four non-dummy models. Each catches 5 of 41 churners when the retention team contacts the top 20 customers by predicted risk. This gives a modest lift over random selection (around 0.6× of the no-info baseline, where contacting 20 random customers would catch around 4 churners on average).

5. Selected Model

The selected candidate is the constrained random forest with hyperparameters `max_depth=4`, `min_samples_leaf=30`, `n_estimators=150`, `max_features="sqrt"` and `class_weight="balanced"`. The reasoning is as follows:

- No model is statistically separable from any other in the comparison set. PR-AUC differences between non-dummy models sit within around 0.03 on the hold-out, well inside the noise envelope at 41 positives.
- Recall at top 10% is tied at 0.122 across the four non-dummy models. The choice therefore favours practical alignment with the retention workflow.
- The random forest is interpretable, stable across random seeds (see section 9), and supports class weights natively.

Gradient boosting (tuned) would also be a defensible choice, since it has higher hold-out PR-AUC and lower Brier score in its baseline form. I am keeping random forest for consistency with the Task 1 indication that tree-based models with controlled depth are preferable at this sample size.

Note on selection validity

The hold-out set informed this selection decision. The reported hold-out numbers should therefore be read as validation evidence with the standard caveat that the selection process touched them. A proper Phase 3 evaluation would hold out a separate test set after model selection is fixed.

		Selected Random Forest at Threshold 0.50 Precision = 0.19 Recall = 0.24 F1 = 0.22	
		Predicted Retained	Predicted Churned
	Actual Retained	117	42
	Actual Churned	31	10

Figure 3. Confusion matrix for the selected random forest at the default threshold of 0.50.

At the 0.50 threshold the model flags 52 customers, of whom 10 actually churn, with 31 of 41 churners missed. The default threshold is not well-suited to retention work, and the next section sets out the trade-off.

6. Threshold and Top-k Trade-off

The retention team will care about the relationship between contact volume and churners captured. The table below sets out the trade-off across both top-k policies and threshold cuts.

Strategy	Customers contacted	Churners caught	Precision	Recall
Top 5%	10	3	30.0%	7.3%
Top 10%	20	5	25.0%	12.2%
Top 15%	30	8	26.7%	19.5%
Top 20%	40	10	25.0%	24.4%
Top 25%	50	10	20.0%	24.4%
Threshold 0.40	179	37	20.7%	90.2%
Threshold 0.45	129	29	22.5%	70.7%
Threshold 0.50	52	10	19.2%	24.4%
Threshold 0.55	12	3	25.0%	7.3%
Threshold 0.60	1	0	0%	0%

Table 2. Outreach trade-off on the hold-out set (200 customers, 41 churners).

Reading the table: a top-5% policy contacts 10 customers and catches 3 churners (precision 30%, recall 7%). A top-20% policy contacts 40 customers and catches 10 churners (precision 25%, recall 24%). Threshold 0.40 sweeps very wide with 179 customers contacted and 37 churners caught (90% recall, 21% precision), which is essentially "contact most of the base". The right operating point depends on the cost of outreach against the value of a saved customer, which is a business decision for the retention team.

7. Probability Calibration

The uncalibrated random forest has a Brier score of 0.234, which sits above the dummy baseline's 0.163. The predicted probabilities sit in a narrow band centred on roughly 0.45. The model can rank customers, while the probability levels themselves are unreliable. This matters when the retention team wants to use scores for budget planning or expected-value reasoning.

Sigmoid (Platt-scaled) calibration was fitted on the training set with internal cross-validation. Sigmoid is safer than isotonic at this sample size, since 163 churners in training is light for the non-parametric isotonic regressor.

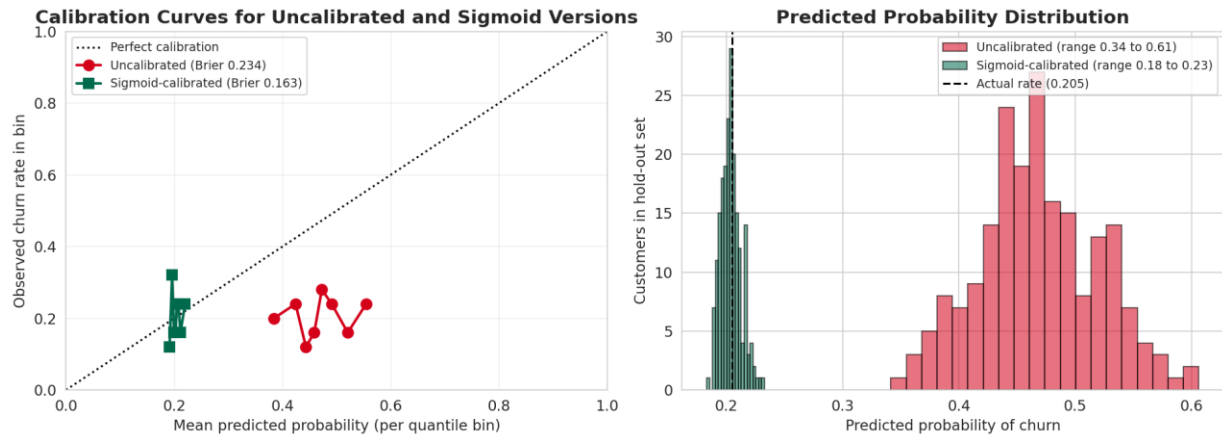


Figure 4. Left: calibration curves for uncalibrated and sigmoid-calibrated versions. Sigmoid-calibrated points sit closer to the perfect-calibration line. Right: predicted probability distributions. The uncalibrated model spreads predictions across 0.34 to 0.61; the calibrated model concentrates them around the actual churn rate (0.205).

Calibration brings the Brier score down to 0.163, matching the dummy baseline. Discrimination changes only slightly (PR-AUC moves from 0.234 to 0.227, within noise) since calibration adjusts probability levels rather than the ranking. For ranked outreach use, either version works. For probability-level decisions, use the calibrated version.

8. Interpretation through Permutation Importance

Feature contribution is measured by permutation importance on the hold-out set, scored on PR-AUC. I have used permutation importance rather than impurity-based importances because Task 1 flagged severe multicollinearity among the transaction features, and impurity rankings scramble badly under that condition. Task 1 noted SHAP as a preference, while permutation importance is the more conservative choice at this sample size. SHAP can be added in Phase 3 once the leakage and tenure data issues are resolved.

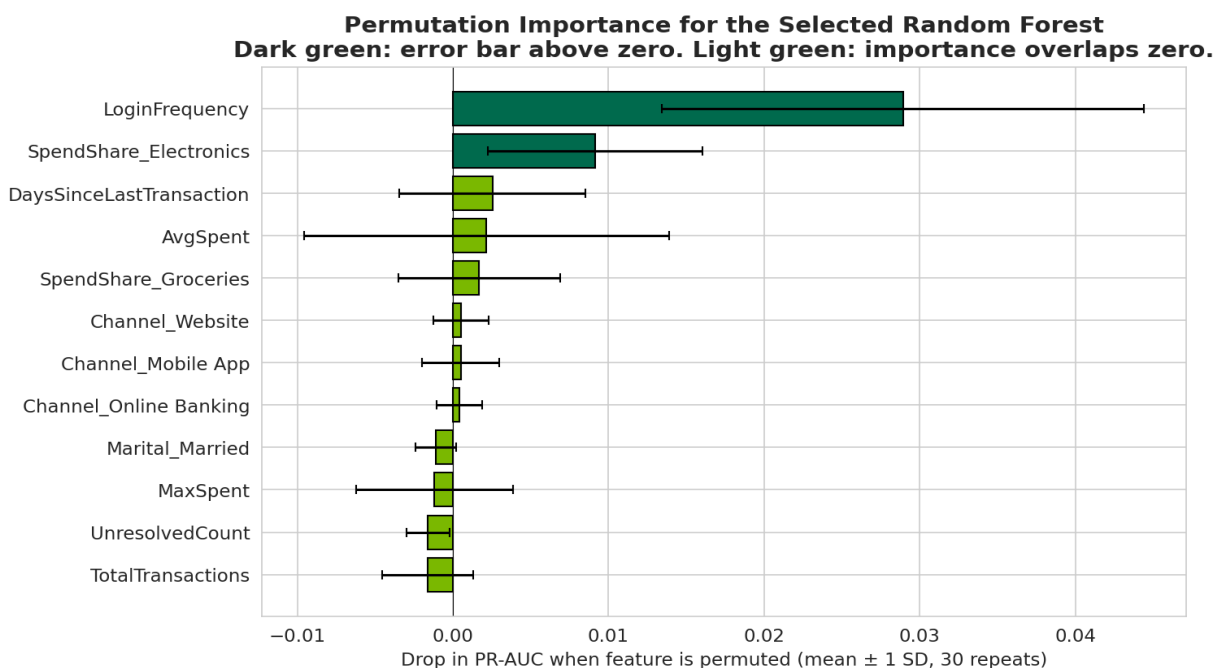


Figure 5. Permutation importance for the selected random forest. Dark green: error bar above zero (clear signal). Light green: importance estimate overlaps zero (noise).

Only LoginFrequency and SpendShare_Electronics have error bars clearly above zero. The ranking of features below those two should be read with caution, since every other feature's 1-SD interval crosses zero. The model leans heavily on LoginFrequency, which is consistent with the EDA and also with the leakage concern raised in the next section.

9. Leakage Sensitivity through Individual Feature Drops

Task 1 flagged LoginFrequency and DaysSinceLastLogin as potentially exposed to target leakage because the online-activity table runs into 2023 while the transaction table ends in 2022 and the churn measurement date is not defined. To narrow down which feature carries the exposure, I retrained the selected model with each feature removed individually and with both removed.

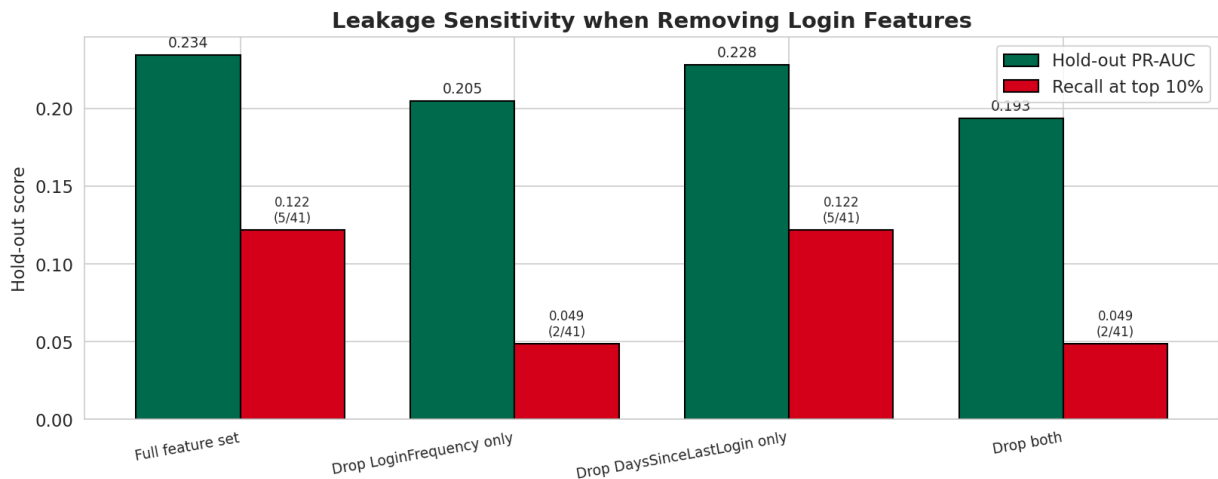


Figure 6. Hold-out performance with login features removed individually and together. Dropping LoginFrequency collapses recall@10% from 0.122 to 0.049; dropping DaysSinceLastLogin alone has almost no effect.

Concentrated exposure on LoginFrequency

The leakage exposure sits firmly in LoginFrequency rather than in DaysSinceLastLogin. Dropping LoginFrequency alone halves the model's business value (recall@10% from 0.122 to 0.049, catching 2 churners instead of 5 in the top 20). Dropping DaysSinceLastLogin alone barely changes anything. The Phase 3 priority is to verify the measurement window for LoginFrequency specifically. DaysSinceLastLogin can probably be retained even if LoginFrequency turns out to be leaky.

10. Subgroup Fairness

Task 1 set out a responsible-use commitment to report model performance by gender, income band and marital status. The table below shows hold-out subgroup metrics. Where a subgroup has fewer than 20 customers or fewer than 5 churners, the metric is marked unstable so that no misleading number gets reported.

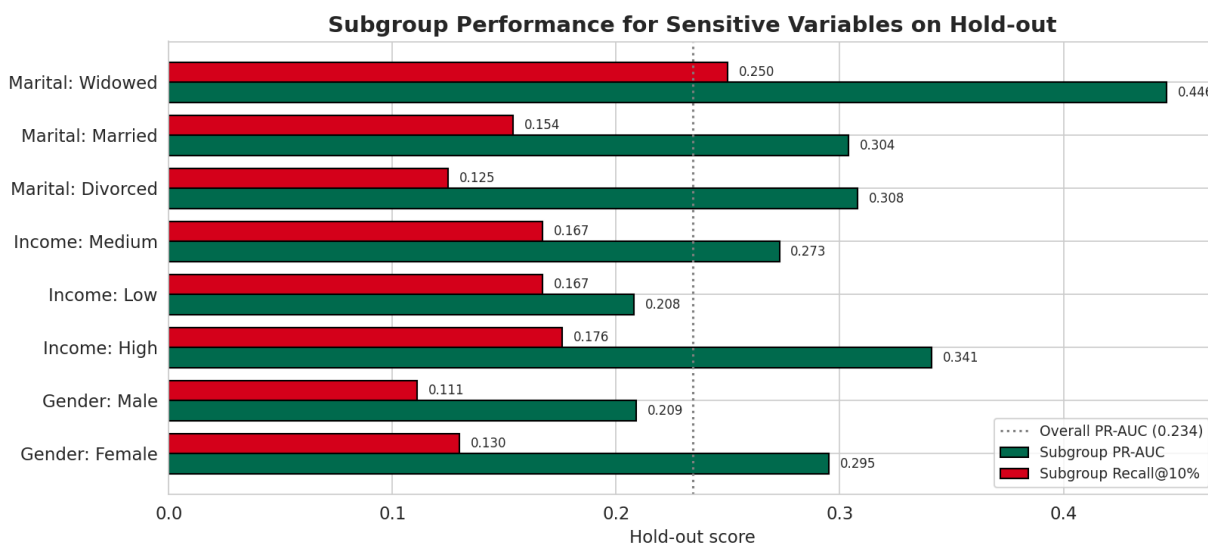


Figure 7. Subgroup PR-AUC and recall@10% on the hold-out set. The grey reference line is the overall hold-out PR-AUC.

Visible gaps:

- Gender: PR-AUC 0.295 (Female) against 0.209 (Male). Recall@10% 0.130 against 0.111.
- Income: PR-AUC 0.341 (High) against 0.208 (Low). The low-income group is the worst served by the model.
- Marital: PR-AUC 0.446 (Widowed) is meaningfully higher than the other groups. Single has only 4 churners, which makes its metrics unstable.

These gaps warrant pre-deployment review. The model is noticeably better at identifying at-risk female and high-income customers, with weaker performance for at-risk low-income customers. A retention intervention driven purely by this model would systematically miss more at-risk customers from the low-income group, which is the wrong direction for a responsible-banking remit. Subgroup catch-rate monitoring should be built into any pilot deployment, with final feature use reviewed alongside legal, compliance and model-risk stakeholders.

11. Business Use and Limitations

The current model is a prototype. The leakage exposure on LoginFrequency, the missing tenure variable, and the subgroup performance gaps all need to be addressed before live use. Within those limits, the model can support:

- A ranked top-k outreach list for low-risk retention actions (service check-ins, app-experience surveys, satisfaction calls), reviewed by the retention team rather than auto-actioned.
- Threshold experimentation in a controlled pilot to see whether the modest lift over random selection translates into measurable retention gains.
- A baseline measurement that future iterations of the model (with tenure, occupation, SME flags and a fixed observation window) can be compared against.

The model should not be used for:

- Pricing decisions or product gating.
- Targeting actions based on demographic variables (Gender_M, IncomeLevel_Ord, Marital_*).
- Budget planning that relies on raw probability levels without first applying the sigmoid calibration provided.
- Customer-facing decisions of any kind without legal, compliance and model-risk sign-off.

12. Recommended Improvements

- Verify the churn measurement date with the project sponsor and rebuild LoginFrequency with a measurement window ending strictly before that date.
- Add tenure, account-opening date, product-holding duration, occupation, SME-account flag, salary-payment marker and relationship-depth fields.
- Extend the customer-service interaction history beyond the current cap of two interactions per customer.
- Re-run the modelling work on a larger sample. With 204 churners, every metric carries a wide confidence interval and feature-importance rankings below the top two are noise.
- Add SHAP per-customer explanations once the data issues above are resolved, so the retention team can see why each customer is flagged.
- Build subgroup catch-rate monitoring into any pilot deployment and review with legal, compliance and model-risk stakeholders before production use.

13. Conclusion

Task 2 produced a complete churn-prediction prototype with proper baselines, hyperparameter tuning, leakage diagnostics and fairness checks. The selected model (a constrained random forest) gives a modest lift over random selection (recall@10% of 0.122 against ~0.049 for random) and remains a prototype rather than a deployment candidate. Three findings sit above the headline metrics:

- Hyperparameter tuning did not improve hold-out performance, consistent with the Task 1 prediction that the dataset is at noise floor.
- The leakage exposure flagged in Task 1 is concentrated in LoginFrequency. Verifying its measurement window is the single most valuable Phase 3 action.
- Subgroup performance varies meaningfully (PR-AUC 0.208 for low-income against 0.341 for high-income). The model cannot be deployed without a fairness review.

The main value of this work is methodological. It sets up a proper modelling pipeline, captures the trade-offs that the retention team will care about, and identifies the data improvements that must come before SmartBank can rely on a churn model in practice.

Deliverables

- Task2_Churn_Model_Report.docx, this report.
- Task2_Churn_Modelling.ipynb, the modelling notebook, which loads the Task 1 cleaned dataset and reproduces every figure, table and number in this report end-to-end.
- selected_churn_model.joblib, the metadata-rich artefact containing the trained model, feature column order, hyperparameters, performance summary, leakage warnings and a do-not-use-for note.
- task2_model_predictions.csv, hold-out predictions with both uncalibrated and sigmoid-calibrated probabilities, RiskRank, and Top10PctFlag for the retention team.